

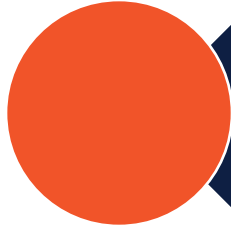
Variability

Computational Statistics Team Teaching
2024/2025

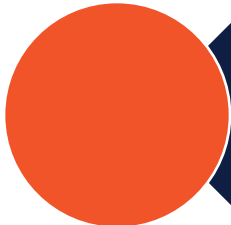
Outlines



What is variability



Measurement of Variability



Shifting dan Scaling



What is variability?

What is spread in data? What is it for?

Variability



Definition

- How much data points differ from the central tendency
- It shows whether the data is **closely packed** or **widely spread out**
- High variability means the data points are far apart, while low variability means they are close together



How to measure variability #1

It can be measured by range, IQR, variance, and std. What are they?

Range

Value of Range

- The **range** is the difference between the **highest** and **lowest** values in a dataset
- The value is affected by the extreme value (too low or too high)

Formula

$$range = x_{max} - x_{min}$$

Range – Example

Math Score Among Students

23 56 45 65 59 55 62 54 85 25

Result

$$\text{range} = 85 - 23 = 62$$

Python

```
1 import numpy as np
2
3 data = np.array([23, 56, 45, 65, 59, 55, 62, 54, 85, 25])
4 data_max = max(data)
5 data_min = min(data)
6 range = data_max - data_min
7 print(range)
```

Quartile

Quartile

A **quartile** is a statistical measure that divides a dataset into **four equal parts**, each containing **25% of the data** when arranged in ascending order. Quartiles help understand the **spread and distribution** of the data.

The Four Quartiles:

1. **Q0 (Minimum Value)** – The smallest number in the dataset.
2. **Q1 (First Quartile / Lower Quartile)** – The **25th percentile**, meaning **25% of the data** is below this value.
3. **Q2 (Second Quartile / Median)** – The **50th percentile**, meaning **50% of the data** is below this value. This is the **median** of the dataset.
4. **Q3 (Third Quartile / Upper Quartile)** – The **75th percentile**, meaning **75% of the data** is below this value.
5. **Q4 (Maximum Value)** – The largest number in the dataset.

Interquartile Range (IQR)

IQR

The **interquartile range (IQR)** measures variability by focusing on the middle 50% of the data. It is the difference between the **third quartile (Q3)** and the **first quartile (Q1)**.

Formula

$$IQR = Q3 - Q1$$

How to find Q3 and Q1? (1)

Case 1

23 25 45 54 55 56 59 62 65 85

Q1

$$Q1 = 45$$

Q2

$$Q2 = \frac{55 + 56}{2} = 55.5$$

Q3

$$Q3 = 62$$

$$IQR = 62 - 45 = 17$$

How to find Q3 and Q1? (2)

Case 2

23 25 45 54 55 55 56 59 62 65 85

Q1

Q2

Q3

$$Q1 = \frac{45 + 54}{2} = 49.5$$

$$Q3 = \frac{59 + 62}{2} = 60.5$$

$$IQR = 60.5 - 49.5 = 11$$

Quartile – The Actual Formula

$$Position = P \times (N + 1)$$

Where P is percentile, N is total number of data

To compute the position of $Q1 \rightarrow 0.25$, and $N = 11$,

$$Position = 0.25(11 + 1) = 3$$

Therefore, the value of $Q1$ lied on the 3rd value in the dataset

How to find the value of Q3 and Q1 using Python?



Python

```
1 import numpy as np
2 import pandas as pd
3
4 data_baru = np.array([23, 56, 45, 65, 59, 55, 62, 54, 85, 25, 55])
5 print(pd.DataFrame(data_baru).describe())
6
7 '''
8 Output
9 count    11.000000
10 mean     53.090909
11 std      17.466852
12 min      23.000000
13 25%      49.500000
14 50%      55.000000
15 75%      60.500000
16 max      85.000000
17 '''
```

Why can python library produce different result?

```
1 import numpy as np
2 import pandas as pd
3
4 data_baru = np.array([23, 56, 45, 65, 59, 55, 62, 54, 85, 25, 55])
5 print(pd.DataFrame(data_baru).describe())
6
7 ...
8 Output
9 count    11.000000
10 mean     53.090909
11 std      17.466852
12 min      23.000000
13 25%      49.500000
14 50%      55.000000
15 75%      60.500000
16 max      85.000000
17 ...
```

Take a look to Q1, Q2, and Q3

When calculating quartiles, the quartile position may **not always fall exactly on a data point**. In such cases, different statistical software, including **Pandas**, may use different interpolation methods to compute the quartile values. By default, Pandas using **LINEAR INTERPOLATION**

Linear interpolation can be calculated from,

$$x_{linear} = i + (j - i) * fraction$$

If you want to use the average between 2 points, in Pandas, you need to define 'midpoint' as interpolation method $\rightarrow (i + j)/2$

Method to computer interpolation in percentile, quantile, and quartile among popular Python library

For Numpy, Pandas, and Scipy

- Linear $\rightarrow i + (j - i) * fraction$ (**default**)
- Lower $\rightarrow$ The value of i
- Higher $\rightarrow$ The value of j
- Nearest $\rightarrow$ The value of i of j depend the nearest
- Midpoint $\rightarrow (i + j)/2$

Anomaly in IQR

The Case

30 40 40 40 40 40 40 40 40 90

Q1

Q2

Q3

$$IQR = 40 - 40 = 0 \quad ???$$



IQR = 0 means the dataset is identical

Variance

Definition

Variance measures how far each value in the dataset is from the **mean**. It is calculated by averaging the squared differences from the mean.

Population

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Sample

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Standard Deviation

Definition

The standard deviation is the square root of the variance. It gives a **measure of spread in the same units as the data**.

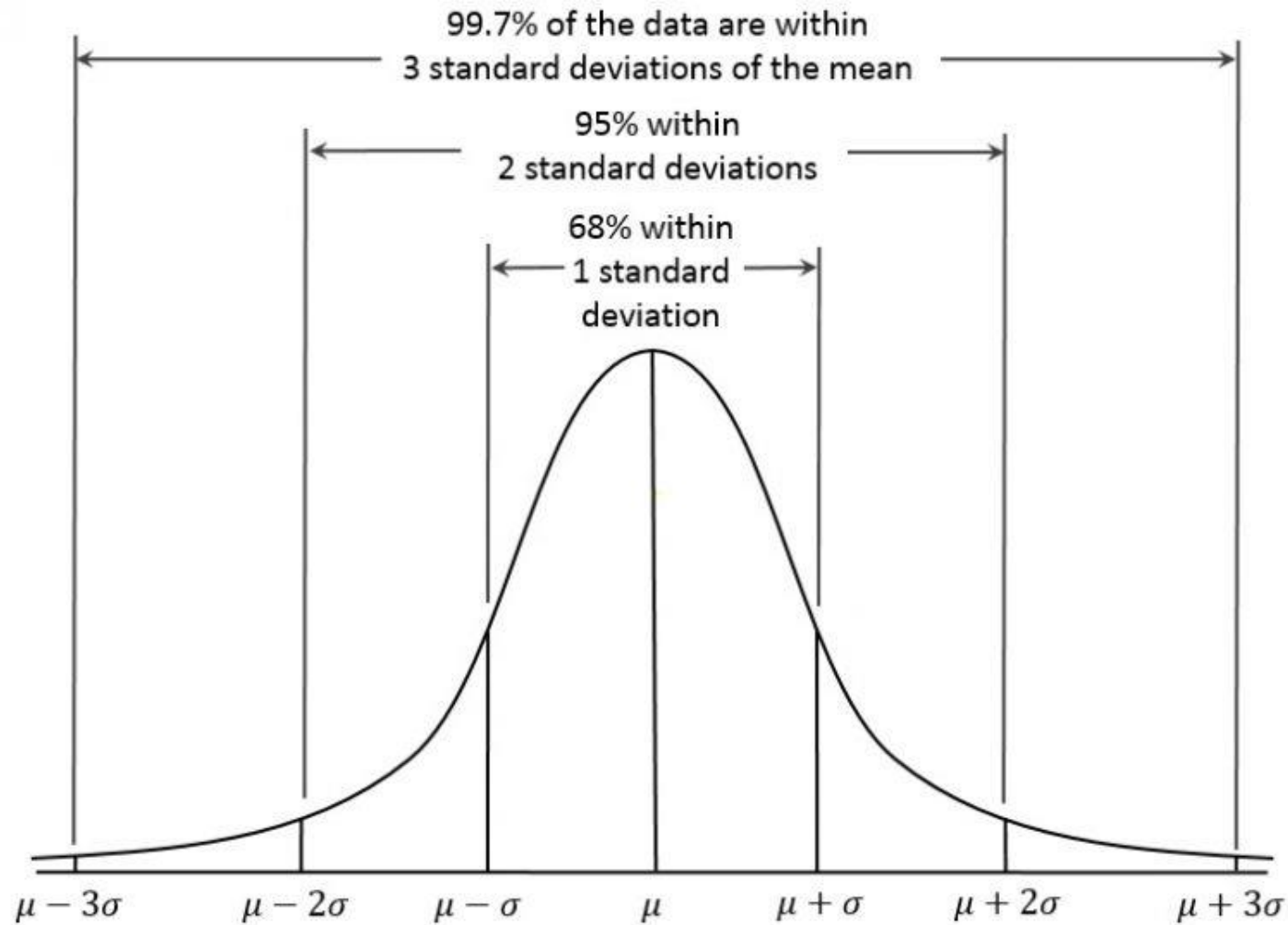
Population

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

Sample

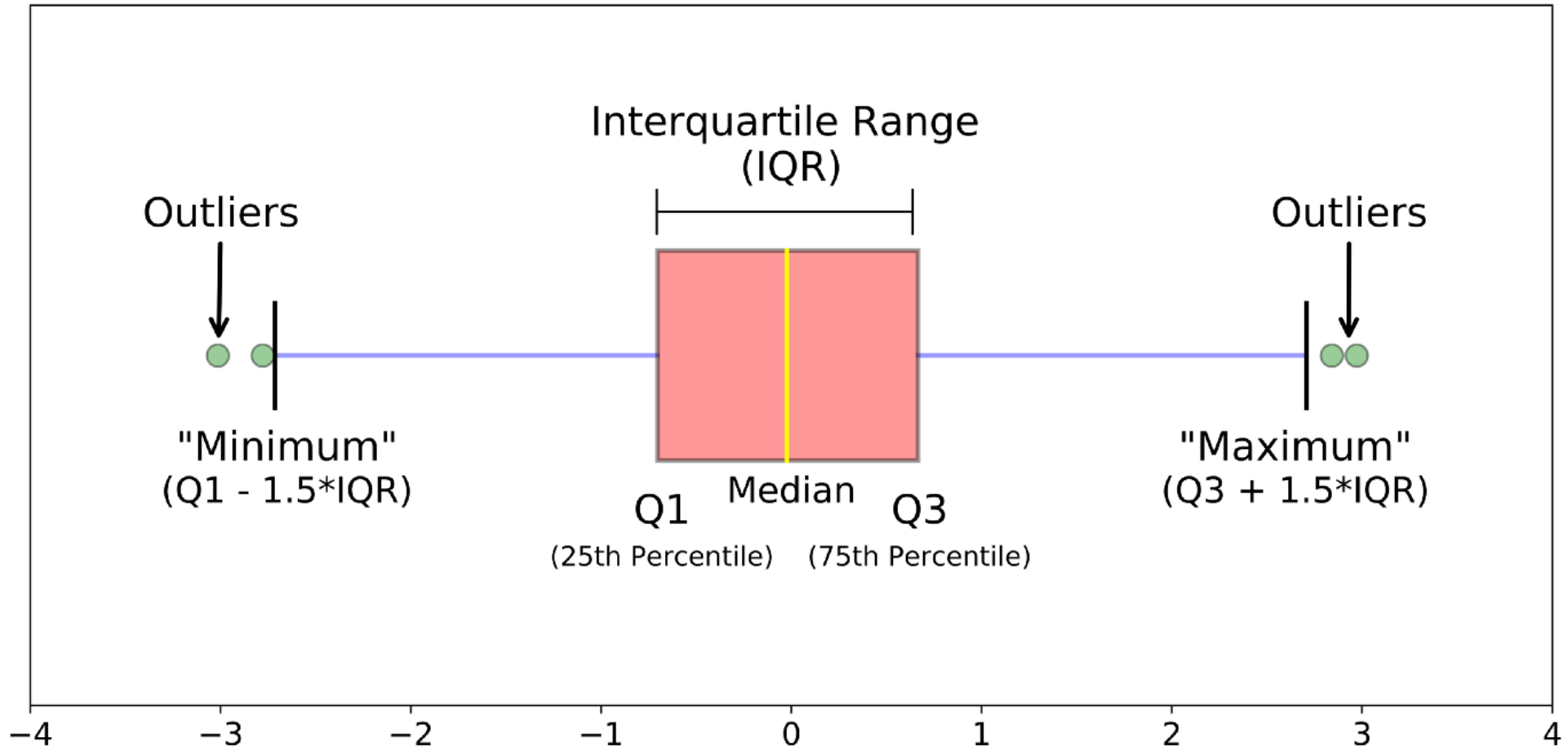
$$S = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Standard Deviation



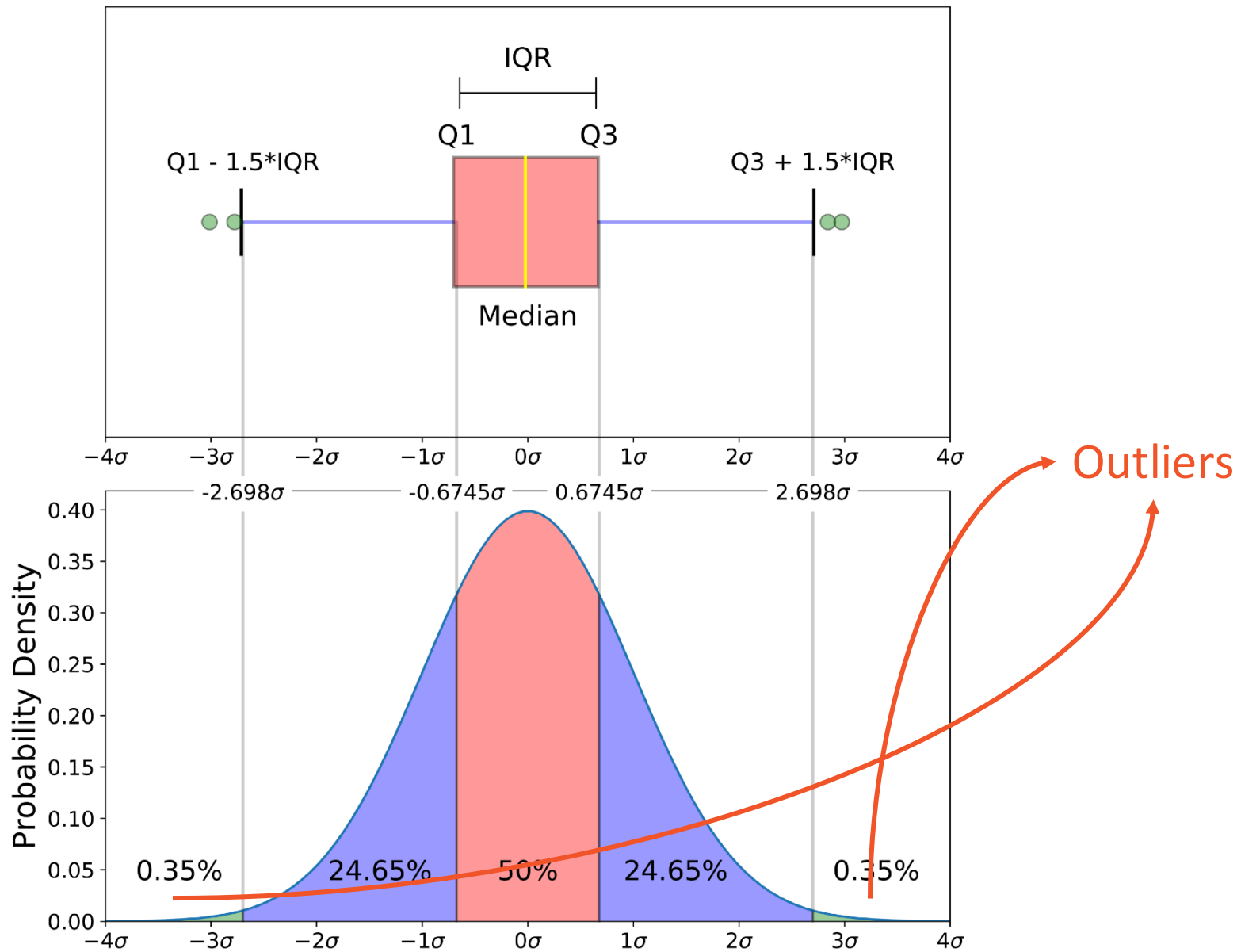
<https://s4be.cochrane.org/blog/2018/09/26/a-beginners-guide-to-standard-deviation-and-standard-error/#:~:text=Standard%20deviation%20tells%20you%20how,standard%20deviations%20of%20the%20mean.>

Spread in Box-Plot



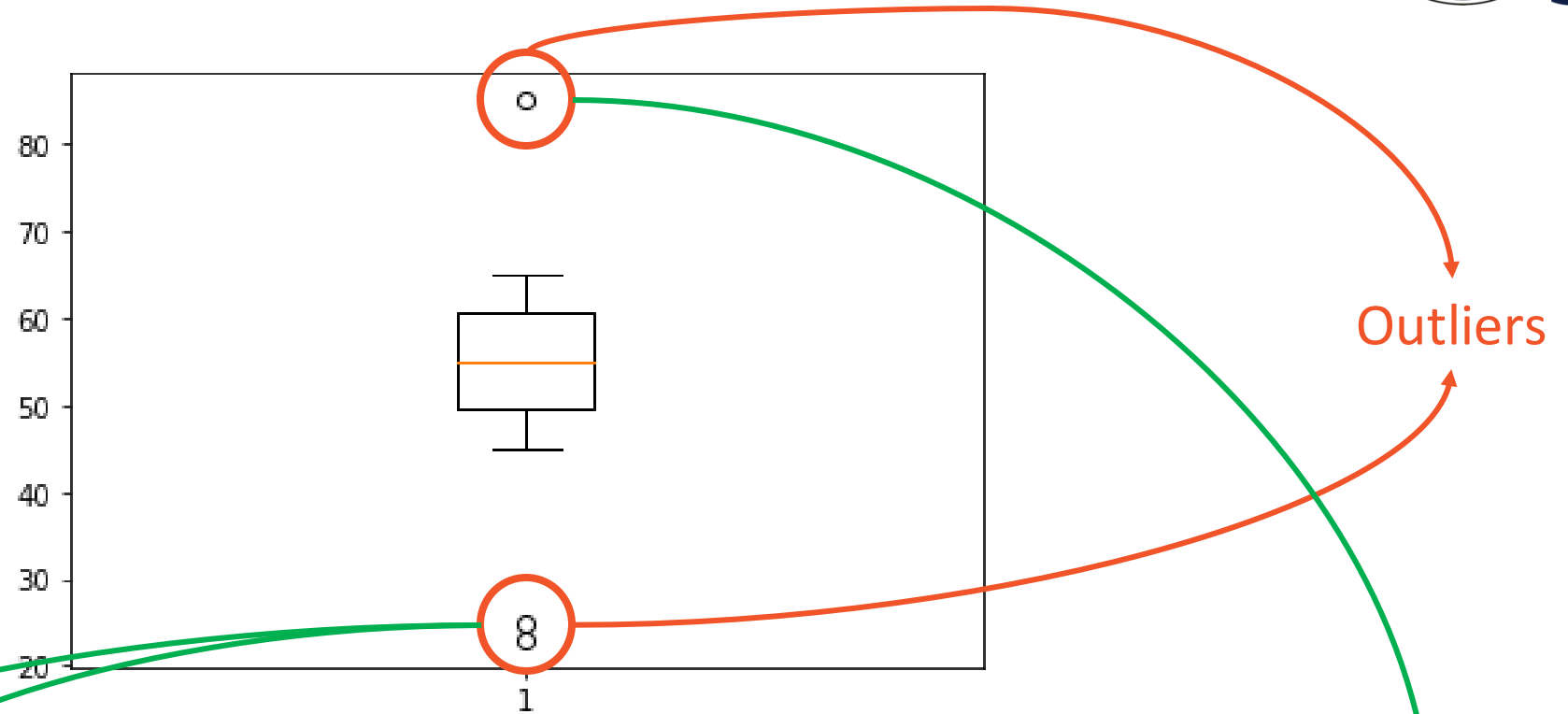
Source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcb51#:~:text=A%20boxplot%20is%20a%20standardized,and%20what%20their%20values%20are.>

Box-Plot in Normal Distribution



Source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcb51#:~:text=A%20boxplot%20is%20a%20standardized,and%20what%20their%20values%20are.>

Another Case in Box-Plot



23

25

45

54

55

55

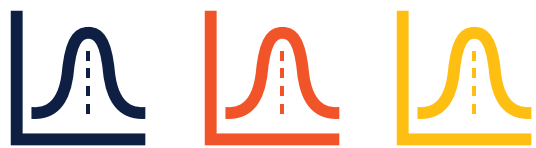
56

59

62

65

85



Shifting dan Scaling

How shifting and scaling affect the central tendency and variability of data

Shifting

What is *shifting*?

Shifting means **adding or subtracting** a constant value to all data points. This operation **changes the mean but does not affect variability (spread) measures** like standard deviation or range.

Exp 1

4, 5, 8, 12 $\rightarrow +2 \rightarrow$ 6, 7, 10, 14

Exp 2

4, 5, 8, 12 $\rightarrow -2 \rightarrow$ 2, 3, 6, 10

The Effect of *Shifting* (1)

OG Data

4, 5, 8, 8, 12

Mean = 7.4

Median = 8

Mode = 8

Range = 8

IQR = 3

STD = 2.8

Shift Data

+2 → 6, 7, 10, 10, 14

Mean = 9.4

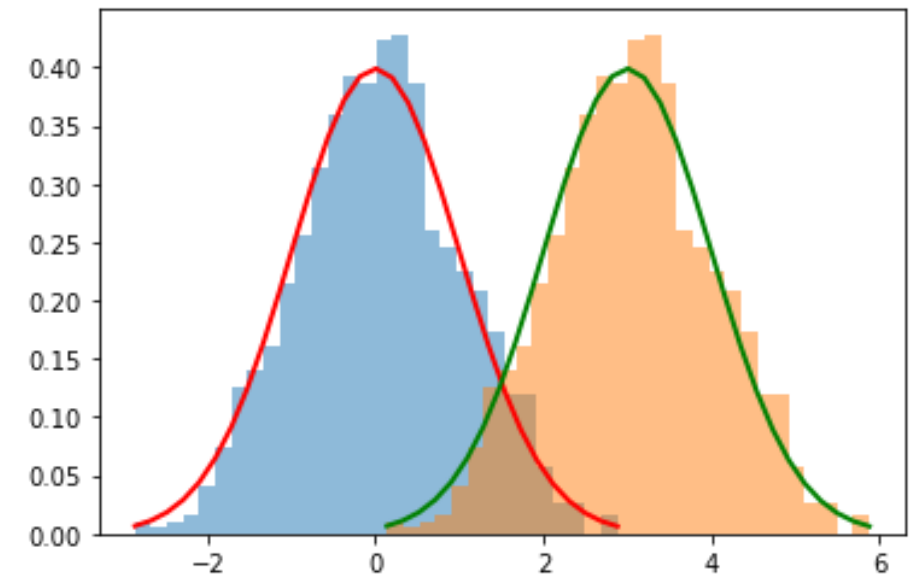
Median = 10

Mode = 10

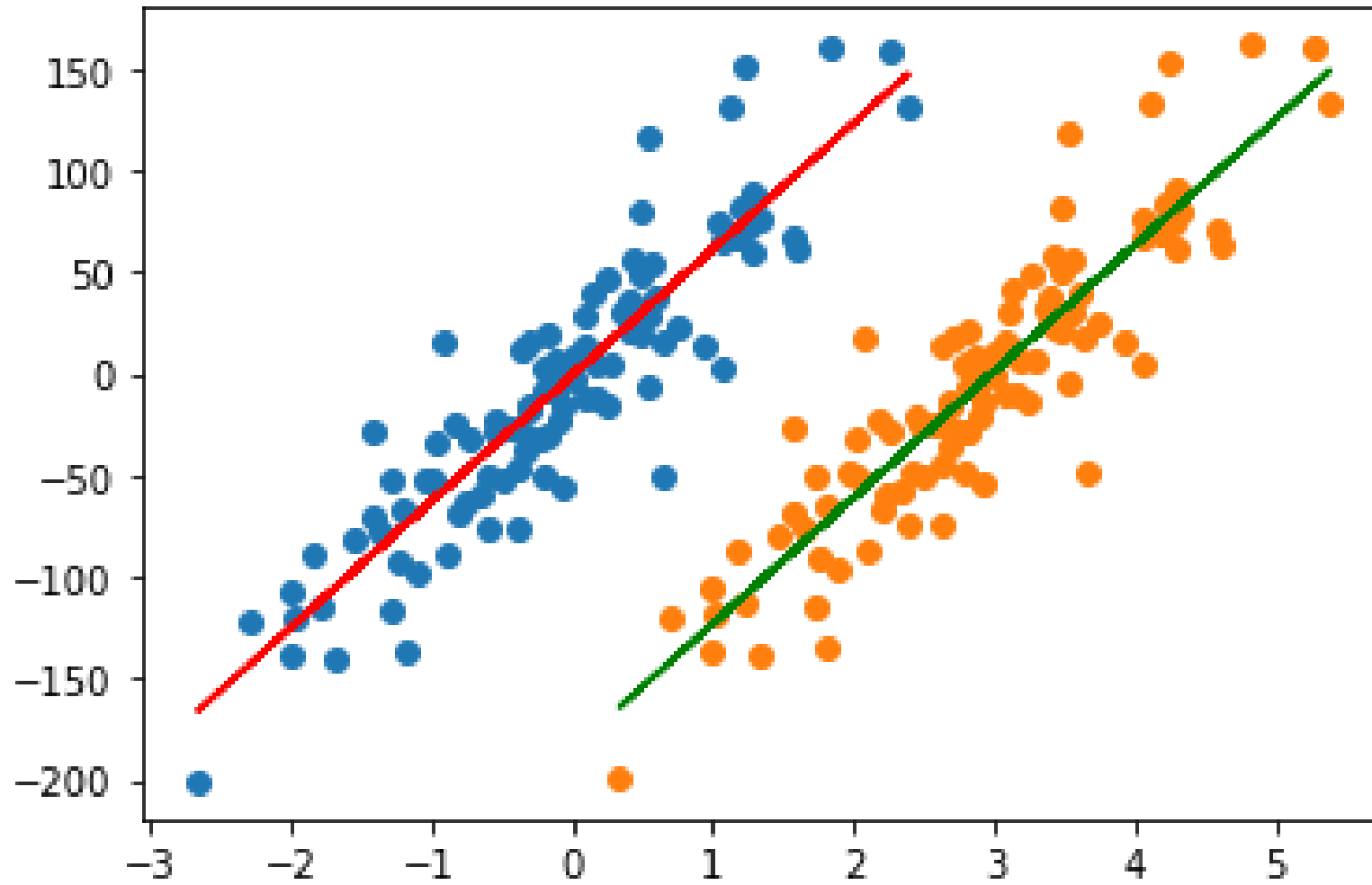
Range = 8

IQR = 3

STD = 2.8



The Effect *Shifting* (2) – Prior Shift in ML



Scaling



What is *scaling*?

Scaling means **multiplying or dividing** all data points by a constant factor. This operation affects both **central tendency** and **variability measures**.

Exp 1

4, 5, 8, 12 $\rightarrow \times 2 \rightarrow$ 8, 10, 16, 24

Exp 2

4, 5, 8, 12 $\rightarrow \div 2 \rightarrow$ 2, 2.5, 4, 6

The Effect *Scaling*

OG Data

4, 5, 8, 8, 12

Mean = 7.4

Median = 8

Mode = 8

Range = 8

IQR = 3

STD = 2.8

Scale Data

$\times 2 \rightarrow 8, 10, 16, 16, 24$

Mean = 14.8

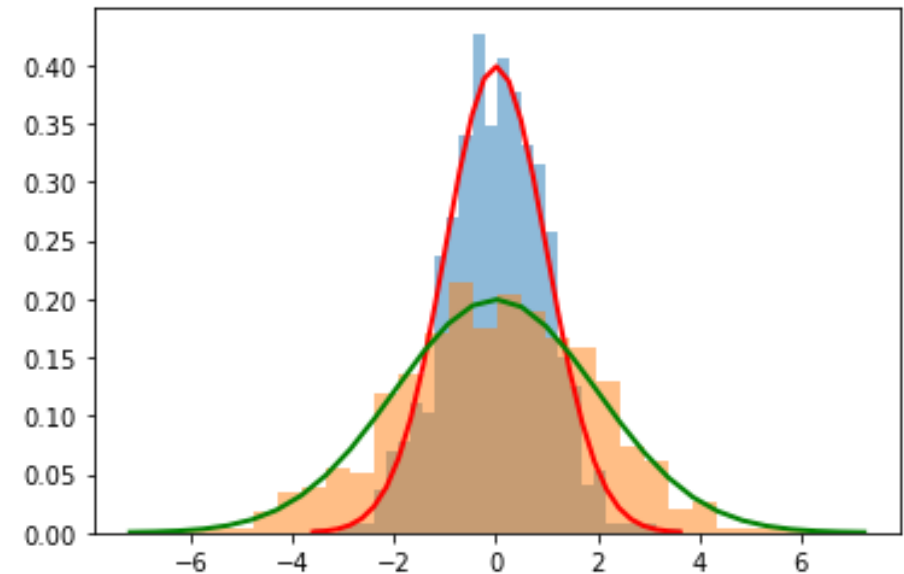
Median = 16

Mode = 16

Range = 16

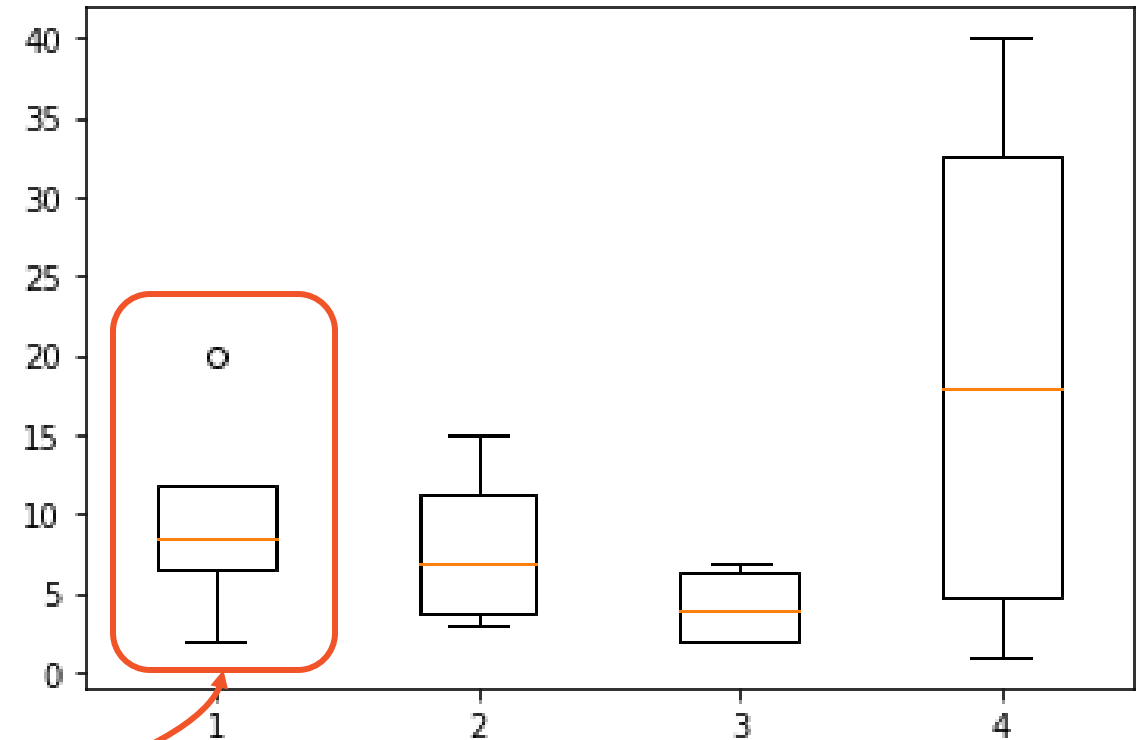
IQR = 6

STD = 5.6



Benefit of Shifting and Scaling – Data Normalization (1)

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 data_raw = np.array([
5     [2, 3, 7, 30],
6     [9, 4, 6, 1],
7     [8, 15, 2, 40],
8     [20, 10, 2, 6]
9 ])
```



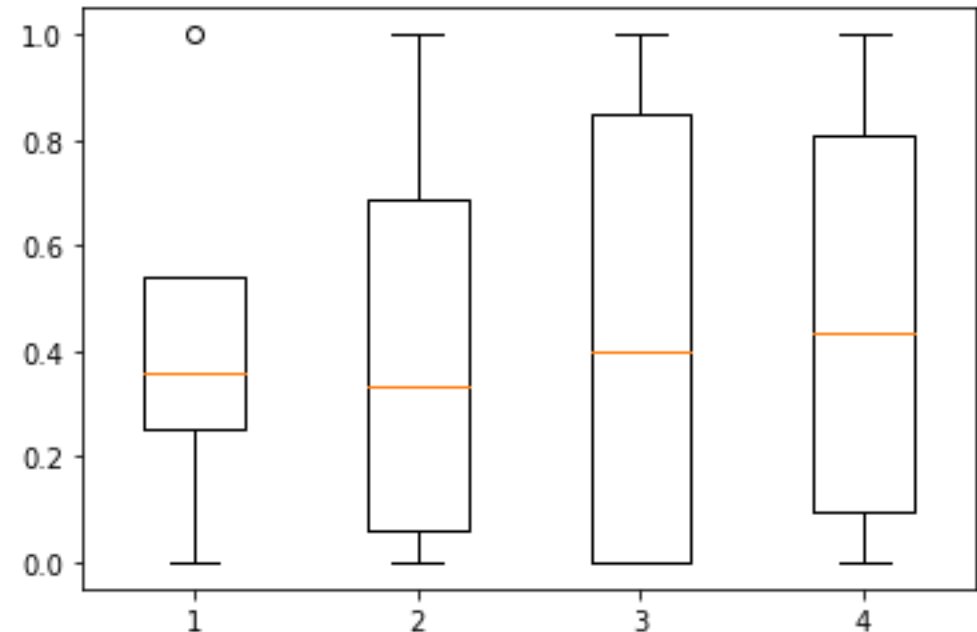
Matplotlib read as columns

Benefit of Shifting and Scaling – Data Normalization (2)



```
1 # Normalisasi Dada - MinMax Scaler
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from sklearn.preprocessing import MinMaxScaler
5
6 data_raw = np.array([
7     [2, 3, 7, 30],
8     [9, 4, 6, 1],
9     [8, 15, 2, 40],
10    [20, 10, 2, 6]
11 ])
12
13 scaler = MinMaxScaler()
14 data_scale = scaler.fit_transform(data_raw)
```

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$



Benefit of Shifting and Scaling – Data Standardization

```
1 # Standarisasi Data
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from sklearn.preprocessing import MinMaxScaler, StandardScaler
5
6 data_raw = np.array([
7     [2, 3, 7, 30],
8     [9, 4, 6, 1],
9     [8, 15, 2, 40],
10    [20, 10, 2, 6]
11 ])
12
13 stand = StandardScaler()
14 data_stand = stand.fit_transform(data_raw)
15
16 plt.boxplot(data_stand)
```

$$x_{stand} = \frac{x - \text{mean}(x)}{\text{std}(x)}$$

